

Inhibitory Precipitation as a Novel Method for Back Extraction of Small Dicarboxylic Acids from Loaded Organic Phase

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Abstract

Reactive extraction is widely regarded as a promising route for the recovery of biobased dicarboxylic acids, but its broader application is constrained by the limitations of back-extraction, particularly product dilution, salt formation, and the need for a subsequent acidification step. In this study, a new back-extraction concept, termed *inhibitory precipitation*, was investigated using itaconic acid as a model small dicarboxylic acid. The method exploits the strong contrast in the solubility of dicarboxylic acids in organic solvents in the presence and absence of extractant. In this approach, dry HCl gas is contacted directly with a trioctylamine (TOA)-loaded organic phase, where it protonates the extractant, suppresses acid-amine complexation, and causes the displaced carboxylic acid to precipitate directly as the free acid. The experimental results demonstrate that inhibitory precipitation is a feasible new mode of back-extraction for reactive extraction systems. The onset of precipitation was predicted well from the solubility behaviour of itaconic acid in TOA-diluent mixtures, indicating that the process is governed primarily by the progressive loss of free TOA available to solubilise the acid. The stoichiometric response during back-extraction, however, was found to depend strongly on the extraction system and could deviate from that expected from simple TOA-diluent solubility behaviour alone. FTIR confirmed that the recovered solid was itaconic acid of high purity, while NMR showed only limited impurity formation in the organic phase during HCl treatment. Among the diluents studied, propyl acetate provided the best overall balance of extraction efficiency, near-1:1 back-extraction stoichiometry, viscosity, and solvent stability. These findings establish a proof of concept for direct recovery of the protonated acid without an aqueous stripping phase and suggest a new route towards more integrated and potentially closed-loop recovery processes for biobased dicarboxylic acids.

1 Introduction

The rapidly growing demand for petrochemical products continues to draw down finite fossil resources and intensifies the need for routes to chemicals from renewable carbon[1, 2]. Within this landscape, dicarboxylic acids are especially attractive as their bifunctional chemistry makes them versatile building blocks for polymers, while also supporting applications in food, cosmetics, pharmaceuticals, and fine chemicals[3–6]. Reflecting their strategic importance, dicarboxylic acids make up nearly half the list of the U.S. Department of Energy’s “Top Value Added Chemicals from Biomass” platform list[7].

Despite this promise, commercial deployment has repeatedly fallen short of projected market potential. Itaconic acid, for example, has historically shown a sizeable gap between installed capacity (on the order of 8×10^4 t y⁻¹) and realised output (reported around 4.14×10^4 t y⁻¹ in the early 2010s)[8–10]. A similar disconnect is seen for succinic acid where in 2007, the market for chemicals derived from succinic-acid intermediates was projected to reach roughly USD 15 billion, yet this growth did not materialise[11, 12]. These examples reflect a broader pattern across biobased dicarboxylic acids where large projected opportunities have been constrained by limited adoption, with production cost as the dominant barrier.

While purely chemical syntheses are possible, they of-

ten depend on expensive or petro-based feedstocks and energy-intensive steps[13–15]. As a result, fermentative production is frequently preferred on sustainability grounds[16, 17]. However, fermentation introduces a structural limitation where the accumulation of acidic products inhibits fungi growth and product formation[18, 19]. Consequently, many dicarboxylic acids exit the bioreactor at modest titers, shifting a disproportionate share of total manufacturing cost to downstream recovery and purification[20, 21].

Downstream isolation is therefore a central economic challenge, commonly accounting for ~ 20 –40% of total production costs and undermining market competitiveness [22, 23]. Conventional recovery relies on direct crystallisation, typically preceded by energy-intensive multistage evaporation[24], or on calcium hydroxide precipitation followed by acidification with excess sulphuric acid which generates calcium sulphate sludge as a waste stream[25]. Alternative routes such as sorption and membrane-based separations (pressure-driven or electrodialysis) can reduce thermal duty but are difficult to scale and are vulnerable to fouling in solids- and organics-rich broths[26–35]. Among candidate technologies, reactive extraction is widely viewed as a particularly promising primary-recovery step because it offers high selectivity, is compatible with scalable liquid-liquid contactors, and can be integrated with continuous *in situ* product re-

covery (ISPR)[24, 36]. By continuously removing acids as they form, ISPR can mitigate accumulation-driven inhibition and improve effective productivity[37–39].

Reactive extraction relies on reversible formation of an acid–extractant complex, in which the extractant selectively binds carboxylic acids and transfers them into an organic phase. Recovery of the acid and regeneration of the extractant therefore require a second step, commonly termed back-extraction, which ultimately determines whether the overall loop can operate continuously and economically[40, 41]. Existing back-extraction approaches span pH, temperature or solvent swings, and ion-exchange stripping, each with distinct trade-offs. In practice, the most valuable back-extraction schemes are those that (i) return the product in its protonated form, (ii) intensify the product stream or, ideally, precipitate it directly, (iii) regenerate the extractant to enable closed-loop recycle, and (iv) minimise waste and auxiliary separations. In this work, we propose and evaluate a novel back-extraction scheme designed to satisfy these criteria simultaneously.

1.1 Current Back Extraction Techniques

There are various methods available for the regeneration of the carboxylic acid from the loaded organic phase.

(i) pH Swing with NaOH Solution In base stripping, the loaded organic phase is contacted with an aqueous NaOH solution, effecting a pH swing that deprotonates the acid and drives it into the aqueous phase as its sodium salt. Stoichiometric amount of NaOH is needed for complete recovery and at sufficiently high base concentration, can intensify the aqueous product stream in a manner favourable for downstream crystallisation[41–43]. The principal drawbacks are that the product is obtained as a salt (necessitating subsequent acidification to recover the free acid) and that the overall process generates substantial inorganic salt waste. In addition, strong base limits the choice of diluents since side reactions with ketones and esters can compromise solvent recycle. Despite these limitations, NaOH stripping remains the most widely used method because of its simplicity and robustness.

(ii) Ion-Exchange with Mineral Acid Solution In acid stripping, the loaded organic phase is contacted with an aqueous mineral acid, which can release the product into the aqueous phase in protonated form. While this avoids a separate acidification step, complete recovery often requires more than stoichiometric acid, and reported performance is sensitive to acid type and concentration[23, 42, 44]. At low to moderate concentrations, stripping generally increases with acid strength and concentration, but several studies report diminished recovery at high mineral-acid concentrations once the extractant becomes excessively protonated and the distribution shifts unfavourably[40].

(iii) pH Swing with trimethylamine (TMA) Trimethylamine (TMA) provides an alternative pH-swing route. Contacting the loaded organic phase with aqueous

TMA extracts the acid as a TMA carboxylate, enabling high recoveries and extractant regeneration at near-stoichiometric reagent usage[43, 45]. A key advantage over NaOH is that TMA is volatile, which can allow recovery of the free acid via evaporation-assisted crystallisation while enabling TMA recycle under reduced pressure, avoiding a separate acidification step[23]. Limitations arise from the maximum practical aqueous TMA concentration (typically ~ 2.5 M), which constrains stream intensification, and from the non-trivial solubility of TMA in the organic phase, which can accumulate upon recycle and impair subsequent extraction[46, 47]. Odour and toxicity also necessitate careful handling and may require additional polishing to ensure complete TMA removal from the product stream[43].

(iv) Temperature-swing Temperature-swing back-extraction is a reagent-free approach that exploits the temperature dependence of the complexation equilibrium. Raising temperature favours the dissociation of the acid–extractant complex, allowing hot water to strip the loaded organic phase while recovering the acid in protonated form and regenerating the extractant. Reported performance, however, is highly system-dependent. For propionic, lactic, pyruvic, and itaconic acids, reported stripping efficiencies are typically low (≈ 5 –30% at a 1:1 water:organic ratio)[41, 43, 48, 49]. In contrast, temperature swing can be substantially more effective for 3-hydroxypropionic acid, with recoveries approaching 80% at 90 °C[50].

(v) Diluent swing Diluent swing introduces a secondary organic component with poor solvating power for the extracted acid and/or reduced ability to stabilise the acid–extractant complex (often termed an antisolvent). The resulting change in organic-phase composition lowers the effective extraction strength by shifting the complexation equilibrium, and promotes release of the acid into an aqueous phase in protonated form while regenerating the extractant. The principal drawbacks are the additional separation steps required to recover and recycle the antisolvent. Liquid antisolvents typically impose a distillation duty, whereas gaseous antisolvents generally require operation under elevated or varying pressure to achieve sufficient dissolution and subsequent removal[23, 51]. In many systems, diluent-swing methods also deliver lower stripping efficiencies than pH-swing approaches.

1.2 Inhibitory Precipitation

Small dicarboxylic acids typically exhibit very low solubility in nonpolar organic solvents because they self-associate strongly through hydrogen bonding and remain highly polar even in their protonated form[52, 53]. In reactive extraction, appreciable transfer to the organic phase therefore occurs primarily through formation of a hydrophobic acid–extractant complex. The central idea of this work is to liberate the bound dicarboxylic acid by introducing hydrogen chloride, which preferentially protonates the extractant and forms a more stable ion pair due to its significantly lower pK_a .

A key distinction from conventional mineral-acid strip-

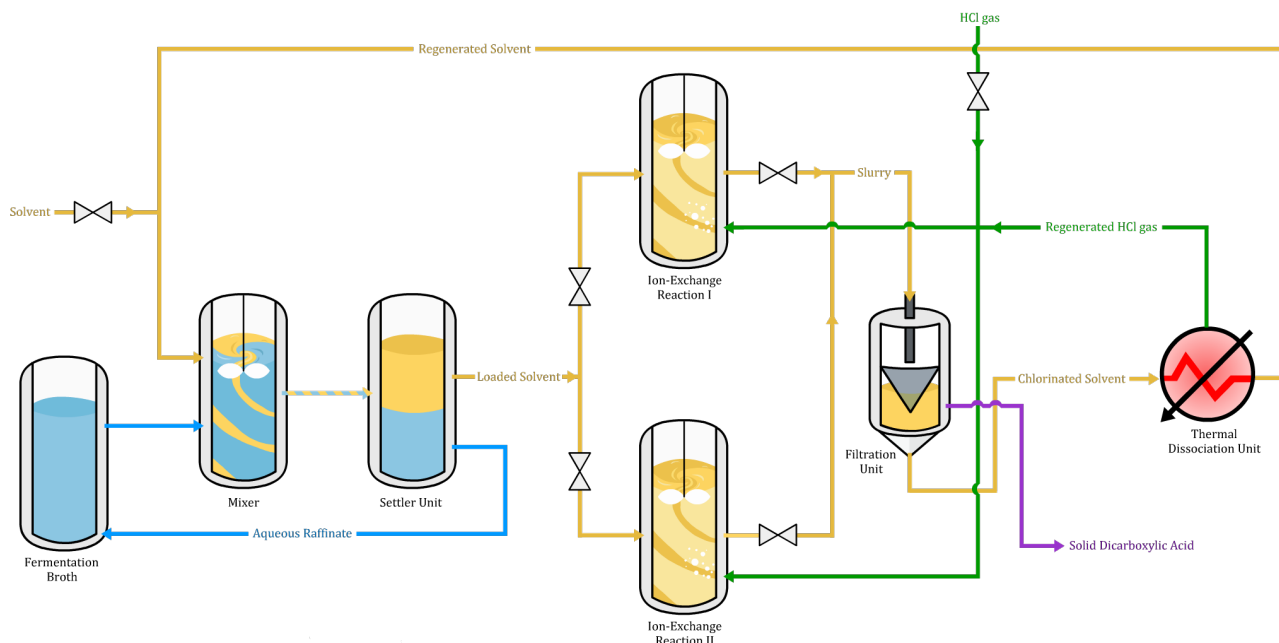


Figure 1: Conceptual process flow diagram of the proposed reactive extraction–inhibitory precipitation process for recovery of small dicarboxylic acids. The aqueous feed is contacted with the organic solvent in the extraction stage, producing an aqueous raffinate and a loaded organic phase. The loaded solvent is then sent to the inhibitory precipitation stage, where dry HCl gas protonates the extractant, suppresses acid–amine complexation, and displaces the dicarboxylic acid as a solid precipitate. The resulting slurry is separated to recover the solid product, while the HCl-treated organic phase is regenerated to produce recycled solvent and regenerated HCl for reuse in the process.

ping is that HCl is introduced as a dry gas rather than as an aqueous acid solution. This changes the thermodynamic behaviour of the system in two important ways. First, in aqueous acid stripping the presence of a water phase creates an additional partitioning equilibrium for both the carboxylic acid and the mineral acid, so excess acid is often required to overcome buffering and distribution effects and to achieve high recovery. In contrast, dry-gas dosing delivers HCl directly to the organic phase and avoids dilution into an aqueous medium, so the displacement reaction is not penalised by interphase partitioning of the stripping reagent. Second, because no aqueous receiving phase is present, the displaced dicarboxylic acid cannot simply repartition into water. Instead, once liberated in its protonated form, it remains in an organic environment where its solubility is intrinsically low, favouring supersaturation and precipitation. The back-extraction step therefore produces the product directly as a solid acid.

After displacement, the organic phase contains the extractant predominantly in its hydrochloride form. For tertiary amine extractants such as trioctylamine, this provides an additional advantage where upon heating, the amine hydrochloride can release HCl as a gas, regenerating the free amine and enabling recycle of both the organic phase and the stripping reagent[54–57]. In principle, this establishes a closed-loop back-extraction scheme that combines direct precipitation of the free acid with regeneration of the extractant and potential recovery of HCl for reuse.

Overall, HCl competitively binds the extractant, displacing the product acid and forcing it to precipitate as the free acid. We term this back-extraction concept

inhibitory precipitation, by analogy to competitive inhibition in enzymatic systems.

2 Methodology

2.1 Materials

Itaconic acid (IA; $\geq 99\%$, Sigma-Aldrich, Schnelldorf, Germany) and trioctylamine (TOA; 98%, Sigma-Aldrich, United States) were used as received. The diluents investigated were 1-octanol (99%, Sigma-Aldrich, Schnelldorf, Germany), 2-octanone (98%, Sigma-Aldrich, China), octane (99%, VWR, France), di-*n*-butyl ether (99%, Thermo Scientific, United Kingdom), and *n*-propyl acetate (99%, Sigma-Aldrich, Schnelldorf, Germany).

Dry hydrogen chloride gas was generated *in situ* from concentrated sulphuric acid (96%, VWR, Leuven, Belgium) and anhydrous sodium chloride ($\geq 99\%$, Honeywell, Germany). Reagents used for analysis included sodium hydroxide (99%, Fluorochem, United Kingdom), hydrochloric acid (5 mol L^{-1} , Fisher Scientific, Austria), silver nitrate standard solution (0.1 mol L^{-1} , Titrisol®, Sigma-Aldrich), nitric acid (68%, Sigma-Aldrich, Schnelldorf, Germany), ammonium iron(III) sulphate (99%, Sigma-Aldrich, Spain), potassium thiocyanate (99%, Sigma-Aldrich, Darmstadt, Germany), and deuterated chloroform (CDCl_3 , 99.8%, VWR, Leuven, Belgium).

2.2 Experimental Procedures

Reactive extraction experiments were performed at 293 K by contacting equal volumes of organic and aqueous phases. The organic phase (25 mL) consisted of TOA at the desired concentration dissolved in the selected

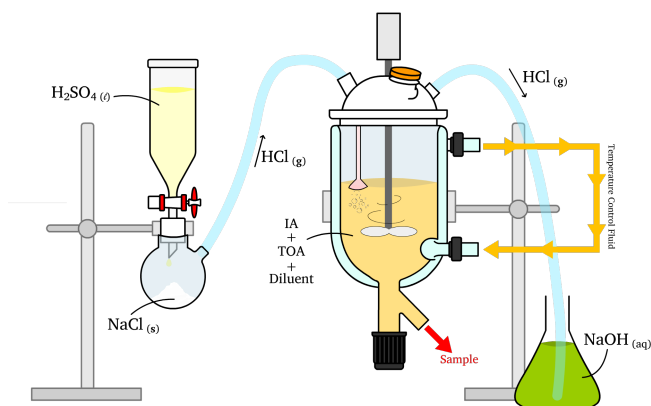


Figure 2: Schematic of the inhibitory precipitation (dry-gas HCl) back-extraction setup. HCl is generated *in situ* by controlled addition of concentrated H_2SO_4 to dry NaCl in a round-bottom flask fitted with a dropping funnel. The gas is transferred through PTFE tubing into a sealed, jacketed reactor containing the loaded organic phase under stirring at 0°C . The reactor outlet is connected to a cold aqueous NaOH scrubber to neutralise off-gas.

diluent. The aqueous phase (25 mL) was an IA solution at 0.615 mol L^{-1} . The phases were mixed vigorously for 2 h, then allowed to fully disengage. The equilibrium aqueous phase was sampled for IA analysis. The amount of IA transferred to the organic phase was calculated by mass balance using the initial and equilibrium aqueous concentrations.

The densities of the pure liquids and their binary mixtures were measured at 293.15 K using an Anton Paar digital vibrating-tube densimeter (DMA 4200). The kinematic viscosities of the mixtures were measured at the same temperature using Ubbelohde viscometers, with size 0b used for all systems except 1-octanol, for which a size 1 viscometer was employed. Dynamic viscosity at each mixture composition was then calculated from the product of the measured kinematic viscosity and the corresponding sample density.

IA solubility in TOA–diluent mixtures was determined by saturation. A large excess of solid IA was added to 25 mL of the organic phase (TOA in diluent), and the slurry was stirred at 70°C for 5 h to accelerate dissolution and equilibration. The mixture was then equilibrated at the target temperature by holding it in a thermostated bath at either 293 K (water bath) or 273.15 K (ice bath) for 8 h. After equilibration, the slurry was immediately cold vacuum-filtered and the clear organic filtrate was collected for analysis.

Back-extraction by inhibitory precipitation was evaluated by introducing dry HCl gas directly into loaded organic phases (Fig. 2). Hydrogen chloride was generated *in situ* by dropwise addition of concentrated sulphuric acid to dry sodium chloride in a round-bottom flask using a dropping funnel. The HCl stream was bubbled into 500 mL of loaded organic phase under constant stirring of 400 rpm. The reactor was maintained at $273.15\text{ K} \pm 0.2\text{ K}$ using a Julabo CF31 cryo-compact refrigerated circulator connected to the reactor jacket. Aliquots (10 mL) were

withdrawn every 10 min over 2 h and immediately cold-filtered prior to analysis.

All experiments were performed three times, and the reported values are presented as the mean, with the associated uncertainty given as the standard deviation of the measurements unless otherwise stated.

2.3 Analytical methods

The concentration of IA in the aqueous phase was determined by acid–base titration using 0.1 mol L^{-1} NaOH with phenolphthalein as the indicator.

Organic-phase analytes were quantified after back-extraction into an aqueous phase using a strong-base pH swing. A measured volume of organic sample was contacted with an excess of prechilled 1 mol L^{-1} NaOH to decomplex the acid–amine species and transfer them to the aqueous phase. To minimise HCl loss during this step, both phases were precooled and mixing prior to capping was kept to a minimum.

For IA concentration determination, the separated aqueous phase was neutralised with 1 mol L^{-1} HCl, diluted as required, and analysed by ion-exclusion HPLC using a SUPELCOGEL C-610H column. The mobile phase was 5 mmol L^{-1} H_2SO_4 at 1.0 mL min^{-1} , with the column maintained at 40°C and UV detection at 220 nm . Aqueous phase samples were analysed directly after appropriate dilution. To account for matrix effects arising from salts, calibration standards were matrix-matched at a fixed NaCl concentration of 0.1 mol L^{-1} .

For HCl analysis, the back-extracted aqueous phase was analysed for chloride by the Volhard method. The sample was acidified with 1 mol L^{-1} nitric acid, treated with an excess of 0.1 mol L^{-1} silver nitrate standard solution, and back-titrated against 0.1 mol L^{-1} potassium thiocyanate using ammonium iron(III) sulphate as the indicator.

The identity of the precipitated solids was confirmed by FTIR spectroscopy through comparison with the spectrum of pure IA. To assess whether impurities were formed during the inhibitory precipitation process, organic-phase samples collected during the experiments were analysed by NMR spectroscopy using a Bruker Avance III HD 400 MHz spectrometer. Both one-dimensional ^1H NMR and two-dimensional ^1H – ^{13}C HSQC spectra were acquired using deuterated chloroform as the solvent.

3 Results and Discussion

3.1 Reactive Extraction of Itaconic Acid

Reactive extraction of itaconic acid from an aqueous feed of 0.615 mol kg^{-1} was investigated using TOA as the extractant at concentrations between 0.1 mol L^{-1} and 1.0 mol L^{-1} . Five diluents were examined, namely 1-octanol, 2-octanone, dibutyl ether, octane, and propyl acetate. These solvents were chosen to provide a set of organic phases with broadly similar molecular size but different polarity, thereby allowing the effect of diluent properties on extraction behaviour to be assessed in a systematic manner.

Reactive extraction performance was evaluated in terms

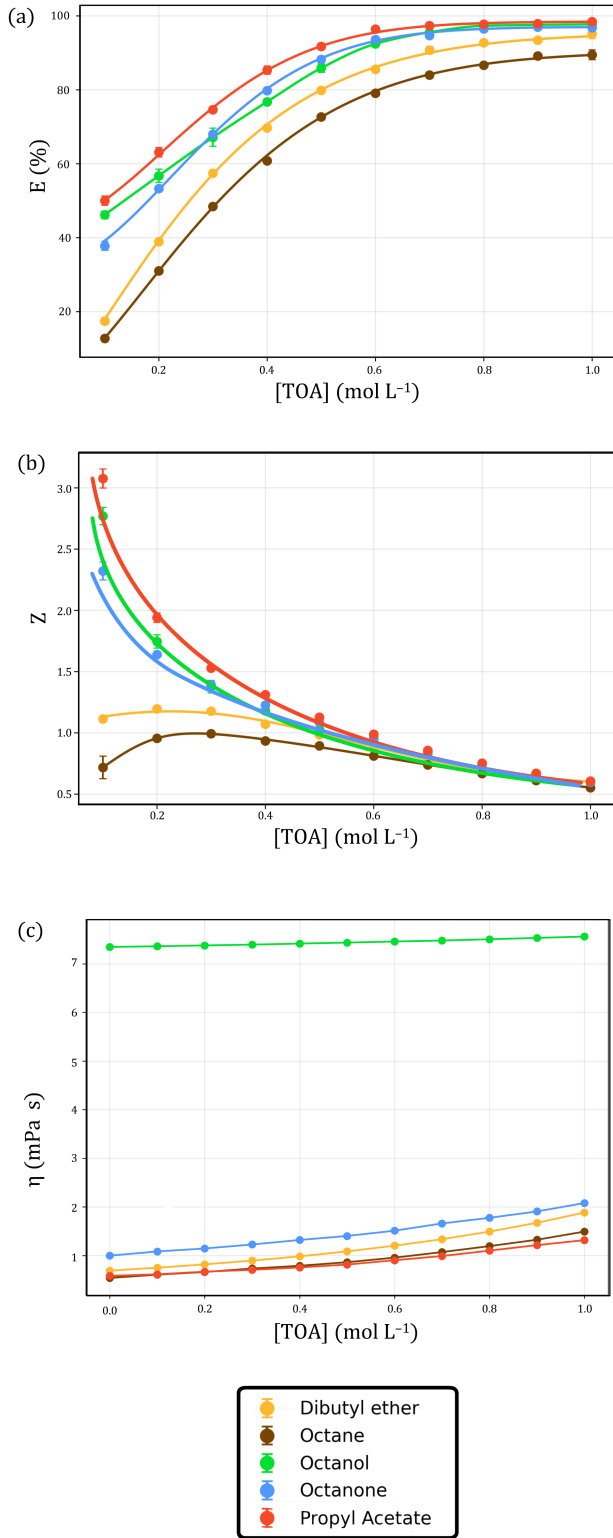


Figure 3: Effect of TOA concentration on the reactive extraction behaviour and physical properties of the investigated diluent systems. (a) Extraction efficiency, E (%), as a function of TOA concentration (mol L⁻¹); (b) loading ratio, Z , as a function of TOA concentration (mol L⁻¹); and (c) dynamic viscosity, η (mPa s), as a function of TOA concentration (mol L⁻¹). The diluents shown are dibutyl ether, octane, octanol, octanone, and propyl acetate. Error bars represent the standard deviation of replicate measurements where visible.

of extraction efficiency, E , loading ratio, Z , and organic-phase viscosity. The extraction efficiency was defined as

$$E = \frac{[\text{IA}]_{\text{org}}}{[\text{IA}]_{\text{aq}}^0} \times 100\% \quad (1)$$

and the loading ratio was defined as

$$Z = \frac{[\text{IA}]_{\text{org}}}{[\text{TOA}]_{\text{org}}^0} \quad (2)$$

where $[\text{IA}]_{\text{aq}}^0$ and $[\text{IA}]_{\text{org}}$ are the itaconic acid concentrations in the initial aqueous phase and in the organic phase after extraction, respectively, and $[\text{TOA}]_{\text{org}}^0$ is the initial TOA concentration in the organic phase.

Figure 3(a) shows that extraction efficiency increased sharply with TOA concentration at low to intermediate concentration, then gradually approached a plateau at higher concentration for all diluents. This behaviour reflects an initial limitation by available amine sites, followed by diminishing gains as the system approaches near-complete extraction. Clear differences between diluents were evident. At 0.6 mol L⁻¹ TOA, where the efficiency curves began to level off, 1-octanol and 2-octanone both achieved extraction efficiencies above 90%, while propyl acetate exceeded 95%. By contrast, dibutyl ether and octane reached only about 86% and 78%, respectively. The superior performance of the more polar diluents is consistent with the well established role of solvent polarity in stabilising acid amine complexes and thereby shifting the extraction equilibrium towards the organic phase. Among the polar diluents, propyl acetate gave slightly higher efficiencies than 1-octanol and 2-octanone over most of the concentration range. Although alcohols and ketones are often reported as particularly favourable media for tertiary amine extraction, the present ranking may reflect the specific composition range investigated here and the use of TOA concentration on a molarity basis rather than the molality basis more commonly adopted in the literature[58, 59].

The loading-ratio data in Fig. 3(b) provide further insight into the nature of the extracted species and help explain the observed phase behaviour. For 1-octanol, 2-octanone, and propyl acetate, the loading ratio was highest at low TOA concentration and then decreased progressively with increasing amine concentration. At the lowest TOA concentration, Z exceeded unity for all three polar diluents, indicating that under acid-rich conditions more than one acid molecule was associated, on average, with each amine molecule in the organic phase[60, 61]. This behaviour is consistent with the formation of higher-order acid amine aggregates when the extractant is the limiting component. As the TOA concentration increased, Z decreased steadily, indicating a gradual transition towards an average stoichiometry close to one acid molecule per amine molecule[62].

A different trend was observed for octane and dibutyl ether. Unlike the more polar diluents, both systems showed an initial increase in loading ratio before declining at higher TOA concentration. In addition, it is observed that both octane and dibutyl ether exhibited

a third-phase formation after contact with the aqueous feed at high TOA concentration, giving a distinct intermediate layer that was presumed to be rich in the TOA–itaconic acid complex[63–65]. These behaviours suggest that acid-rich aggregation was more pronounced in these weakly solvating diluents and is undesirable as they complicate phase disengagement and reduces the practicality of solvent recovery.

Although solvent screening in reactive extraction is often based primarily on equilibrium performance, the viscosity of the organic phase is also a key selection criterion because it directly affects pumping, mixing, and phase disengagement. Fig. 3(c) shows that for all diluents, viscosity increased monotonically with TOA concentration, which reflects the increasing contribution of the extractant to the overall resistance to flow and is consistent with literature data[66]. With the exception of 1-octanol, all systems remained at or below approximately 2 mPa s across the full TOA concentration range in this study. Within this group, viscosity increased in the order octane, dibutyl ether, propyl acetate, and 2-octanone. These differences were clearly distinguishable, but all four diluents remained within a range that is unlikely to impose major constraints on liquid handling or phase-contacting operations. A markedly different behaviour was observed for 1-octanol. Even in the absence of TOA, its viscosity was already far higher than that of the other diluents, and the viscosity of the TOA-octanol mixtures remained substantially above the rest of the solvent set across the entire concentration range. While 1-octanol provided extraction efficiencies comparable to those of 2-octanone and propyl acetate, its much higher viscosity would be expected to hinder pumping, mixing, mass transfer, and phase separation in an operating process.

For the subsequent back-extraction study, only 1-octanol, 2-octanone, and propyl acetate were selected, since third-phase formation in octane and dibutyl ether would complicate operation and phase separation. Back extraction was performed using artificially prepared organic phases formulated to represent the equilibrium composition at 0.6 mol L^{-1} TOA. This concentration was chosen because the extraction efficiency had approached a plateau and the loading ratio was close to unity, indicating efficient use of the extractant with minimal excess free TOA.

3.2 Solubility

The feasibility of inhibitory precipitation depends on the strong difference between the solubility of free IA and IA complexed with TOA in the organic phase. In the absence of extractant, small dicarboxylic acids show only limited solubility in most organic solvents, whereas the presence of a tertiary amine markedly increases acid uptake through acid–amine complexation. For this reason, the solubility of IA in TOA–diluent mixtures is a key parameter for evaluating whether the dissolved acid can later be forced out of solution by suppressing the extracting ability of the amine.

Figure 4 shows that IA solubility increased with TOA concentration in all three diluents at both temperatures. This trend confirms that the dominant contribution to IA solubilisation is the formation of TOA-containing complexes that remain soluble in the organic phase. The

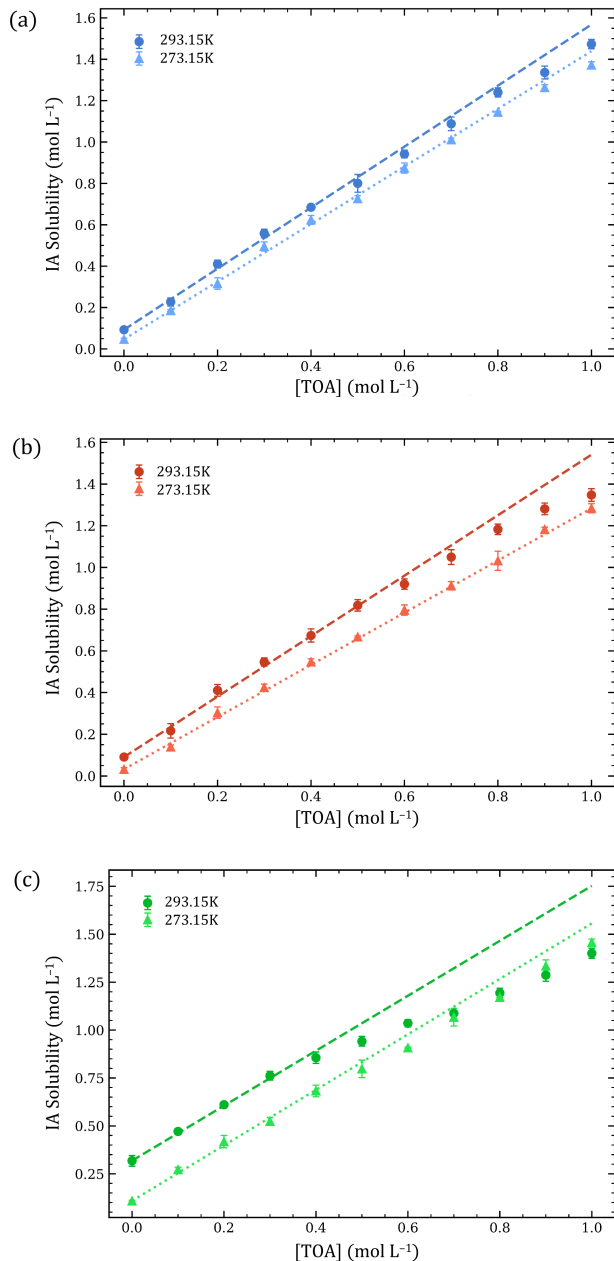


Figure 4: Solubility of itaconic acid in TOA–diluent mixtures as a function of TOA concentration. Panels (a), (b), and (c) correspond to the systems employing 2-octanone, propyl acetate, and 1-octanol, respectively. Symbols represent the experimental data at 293.15 K and 273.15 K. The dashed lines show the linear fits to the 293.15 K data using only the first six points, while the dotted lines show the linear fits to the full 273.15 K dataset. The fitted relations are $y = 1.475x + 0.092$ and $y = 1.392x + 0.046$ for 2-octanone at 293.15 K and 273.15 K, respectively, $y = 1.450x + 0.090$ and $y = 1.251x + 0.032$ for propyl acetate, and $y = 1.434x + 0.318$ and $y = 1.448x + 0.109$ for 1-octanol.

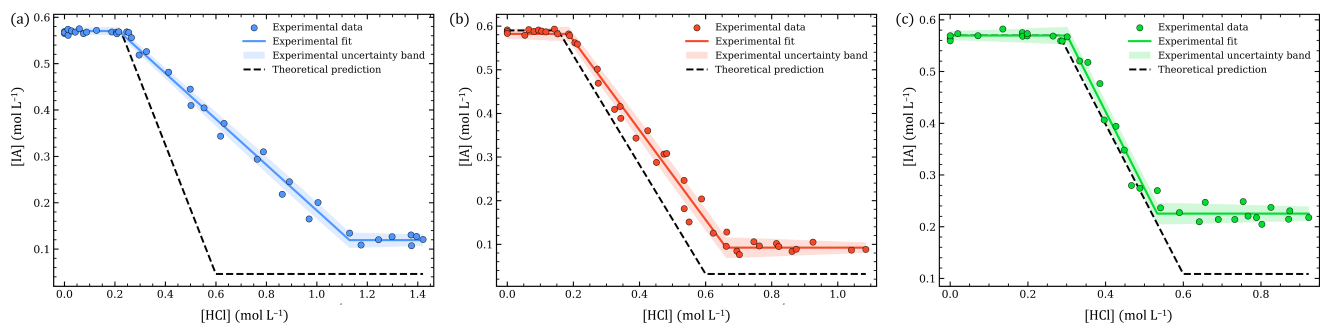


Figure 5: Comparison of experimental inhibitory precipitation behaviour with the theoretical prediction derived from the solubility model for the systems employing (a) 2-octanone, (b) propyl acetate, and (c) 1-octanol at 273.15 K. Symbols represent the experimental data. The solid lines show the fitted experimental trends obtained using a piecewise plateau-linear-plateau model. The shaded bands represent a smoothed local uncertainty estimated from the deviations between the experimental data and the fitted curve using Gaussian kernel smoothing. The dashed lines show the theoretical prediction derived from the low-temperature solubility model.

temperature effect, however, depended strongly on the diluent. In 2-octanone, shown in Fig. 4(a), the difference between 273.15 K and 293.15 K was relatively small over the full concentration range. Propyl acetate in Fig. 4(b) showed a clearer separation between the two temperatures. The largest temperature dependence was observed for 1-octanol in Fig. 4(c), where the intercept at zero TOA was substantially higher at 293.15 K, indicating a much larger intrinsic solubility of IA in the warmer solvent.

At 273.15 K, the solubility curves were essentially linear over the entire concentration range investigated. The fitted slopes were 1.392 for 2-octanone, 1.251 for propyl acetate, and 1.448 for 1-octanol, with corresponding intercepts of 0.046, 0.032, and 0.109 mol L⁻¹, respectively. These intercepts represent the intrinsic solubility of IA in the neat diluents in the absence of TOA and show that 1-octanol dissolves substantially more IA than 2-octanone or propyl acetate even without extractant. A slope close to unity indicates that each additional mole of TOA introduced into the organic phase allows approximately one additional mole of IA to remain dissolved, which is consistent with a dominant minimum stoichiometry close to 1:1. The fact that all slopes were slightly greater than unity suggests that the apparent solubilisation capacity was modestly higher than strict 1:1 association. Such behaviour is consistent with partial overloading or the formation of acid-rich aggregates, which is reasonable for dicarboxylic acids because TOA systems are known to support higher-order acid-amine complexes under some conditions.

A slightly different behaviour was observed at 293.15 K. In all three diluents, the initial low-TOA region remained approximately linear, with slopes of 1.475 for 2-octanone, 1.450 for propyl acetate, and 1.434 for 1-octanol over the fitted range. At higher TOA concentration, however, the experimental points fell below the linear extrapolation. This deviation was minor for 2-octanone, more pronounced for propyl acetate, and strongest for 1-octanol. In the octanol system, the downward curvature at 293.15 K was large enough that the 273.15 K curve overtook the 293.15 K curve at high TOA concentration. The most plausible interpretation is that two competing effects are operating. Raising the temperature increases

the non-reactive solubility of IA in the diluent, which explains the larger intercepts at low TOA concentration. At the same time, the acid amine association that dominates at higher TOA concentration becomes less favourable as temperature rises since the reaction is exothermic.

Although the reduction in IA solubility at lower temperature is most pronounced in 1-octanol, cooling is expected to benefit all diluent systems. Beyond its effect on acid solubility, lower temperature also favours the dissolution of HCl gas in the organic phase, which should enhance gas absorption during inhibitory precipitation. Consequently, operation at reduced temperature may still be advantageous even when the downward shift in the solubility curve is relatively small.

3.3 Inhibitory Precipitation

Figure 5 presents the residual IA concentration in the organic phase as a function of dissolved HCl concentration for the 2-octanone, propyl acetate, and 1-octanol systems. In this form, the inhibitory precipitation data can be viewed as a reversed solubility problem. The addition of HCl progressively suppresses the extracting ability of TOA through protonation, thereby reducing the amount of IA that can remain solubilised in the organic phase. For comparison, the dashed curves show a simplified theoretical prediction derived from the IA solubility data at 273.15 K. This model assumes that the total solubilising capacity is the sum of the intrinsic IA solubility in the neat diluent and the contribution from the remaining free TOA. We assume that each increment of available TOA provides a constant solubilisation capacity, as determined by the slope of the low-temperature solubility isotherms. Precipitation is predicted to occur once the calculated solubility limit falls below the initial IA loading of the organic phase. This three-stage behavior is captured mathematically by the following nested expression:

$$[IA]_{\text{org}} = \min([IA]_{\text{org}}^0, \max(S_0, S_0 + m \cdot [TOA]_{\text{org}}^{\text{free}})) \quad (3)$$

where S_0 is the intrinsic solubility of IA in the pure diluent, m is the slope obtained from the 273.15 K solubility curve. The concentration of free amine available for complexation, $[TOA]_{\text{org}}^{\text{free}}$, is estimated by assuming

a stoichiometric 1:1 displacement by HCl:

$$[\text{TOA}]_{\text{org}}^{\text{free}} = [\text{TOA}]_{\text{org}}^0 - [\text{HCl}]_{\text{org}} \quad (4)$$

Both the theoretical prediction and the experimental data exhibit three characteristic regions. At low HCl concentration, an initial plateau is observed in which the IA concentration remains essentially constant. In this region, the added HCl is accommodated by the organic phase without yet forcing IA out of solution, since sufficient TOA remains available to sustain the original IA loading. This is followed by a descending region in which the IA concentration decreases approximately linearly with increasing HCl concentration. Here, progressive protonation of TOA reduces the solubilisation capacity of the organic phase, and IA is displaced from the amine-containing phase. At still higher HCl concentration, a second plateau is reached, indicating that most of the TOA-bound IA has already been removed and that the remaining IA corresponds to a residual solubility limit in the HCl-treated organic phase.

A notable result is that the onset of precipitation was predicted very well by the simple solubility-based model for all three diluents. Based on the theoretical curves, the initial plateau was predicted to end at approximately 0.22 mol L^{-1} HCl for 2-octanone, 0.17 mol L^{-1} for propyl acetate, and 0.30 mol L^{-1} for 1-octanol. The corresponding experimental break points, estimated from the fitted curves, occurred at about 0.24 mol L^{-1} , 0.20 mol L^{-1} , and 0.31 mol L^{-1} , respectively. The agreement is therefore exceptionally close for all systems and it shows that the critical HCl dosage required to saturate IA removal follows directly from the TOA-dependent solubility behaviour of the corresponding diluent system.

The descending region provides more detailed insight into the back-extraction mechanism. In all three diluents, this region remained approximately linear, but the experimental slopes did not always coincide with those predicted from the 273.15 K solubility curves. For 1-octanol, the agreement was excellent, with an experimental slope of 1.48 compared with a predicted value of 1.45. This close correspondence suggests that IA displacement in octanol follows the expected competitive inhibition mechanism with little additional complexity. Propyl acetate also showed reasonably good agreement, with an experimental slope of 1.03 compared with a predicted value of 1.25. By contrast, 2-octanone exhibited a much shallower response, with an experimental slope of only 0.49 compared with a predicted value of 1.39.

The magnitude of the descending slope is important because it reflects the effective stoichiometry of the back-extraction step with respect to HCl and IA. A slope close to 1 indicates that protonation of TOA displaces approximately one mole of IA for each mole of HCl added, which is consistent with a simple 1:1 response. A slope smaller in magnitude than unity suggests that HCl addition does not translate directly into equivalent IA release. One plausible explanation is that the diluent stabilises higher-order TOA-IA complexes, so that protonation proceeds without immediate disruption of all bound acid species. By contrast, a slope larger in magnitude than unity would imply that protonation of a single amine centre destabilises multiple acid-rich complexes and releases more than one mole of IA on average. The near-quantitative

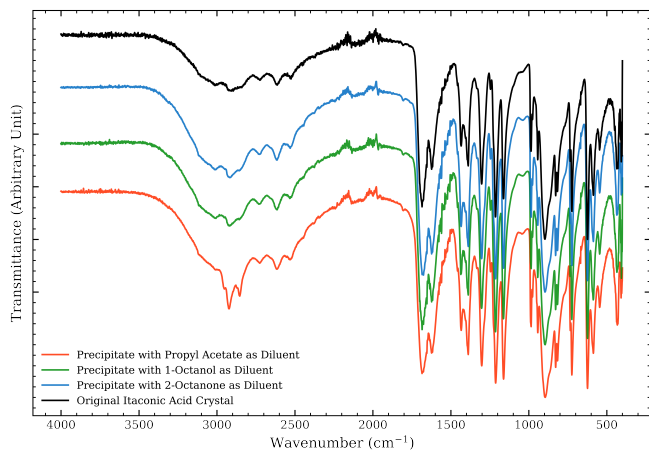


Figure 6: FTIR spectra of the original itaconic acid crystal and the precipitates obtained after inhibitory precipitation using propyl acetate, 1-octanol, and 2-octanone as diluents. The precipitated solids show the same characteristic absorption bands as the reference itaconic acid crystal, including agreement in both the functional-group region and the fingerprint region, confirming that the recovered solid is itaconic acid. The much higher absorbance of the precipitate spectra is attributed to differences in sample packing during measurement.

agreement observed for 1-octanol therefore points to a relatively direct and predictable back-extraction pathway, whereas the much smaller slope in 2-octanone suggests a more complex redistribution of species during HCl addition.

The second plateau also reveals an important limitation of the simple theoretical treatment. In all three cases, the final experimental IA concentration remained substantially higher than the value predicted from the intrinsic solubility term. For 2-octanone, the final plateau was 0.120 mol L^{-1} experimentally, compared with a predicted value of only 0.046 mol L^{-1} . For propyl acetate, the corresponding values were 0.092 mol L^{-1} and 0.032 mol L^{-1} . For 1-octanol, the discrepancy was again pronounced, with an experimental plateau of 0.225 mol L^{-1} compared with a predicted value of 0.109 mol L^{-1} . This shows that the terminal state of the system cannot be described as simple dissolution of IA in the neat diluent alone. After HCl addition, the organic phase no longer consists of only diluent and free TOA, but also contains substantial amounts of protonated amine and associated ionic species. The residual IA solubility is therefore governed by the properties of an HCl-rich, TOA-HCl-containing organic phase rather than by the pure solvent itself. Residual interactions with protonated TOA, ion pairing, or changes in the local solvent environment may thus allow more IA to remain dissolved than predicted by the intrinsic-solubility term alone.

3.4 Purity

The identity of the recovered solid was confirmed by FTIR spectroscopy. Figure 6 compares the spectra of the precipitates obtained from the inhibitory precipitation experiments with that of the original itaconic acid crystal. In the functional-group region, the precipitated samples show the same broad O-H stretching absorption and the

same characteristic carbonyl absorption as the reference material. More importantly, the spectra also exhibit close agreement throughout the fingerprint region, where the band pattern is highly specific to the molecular structure. The strong correspondence across both regions indicates that the precipitated solid is indeed itaconic acid of high purity and that no major chemical transformation occurred during the precipitation step.

To assess the suitability of the organic phase for recycle, it is necessary to determine whether inhibitory precipitation generates persistent side products in solution. Figure 7 shows the time-resolved ^1H NMR spectra of the organic phase during the process, where t_0 corresponds to the initial loaded organic phase before HCl addition, t_1 to a composition near the end of the first plateau, t_2 to a composition near the end of the linear descent, and t_3 to the final stage of the experiment in the second plateau. Across all three diluent systems, the IA resonances at approximately 3.3, 5.6, and 6.0 ppm remain essentially unchanged between t_0 and t_1 , decrease markedly between t_1 and t_2 , and then show only minor further change between t_2 and t_3 . This behaviour mirrors the concentration profiles in Fig. 5 and supports the interpretation that the dominant chemical event observed by NMR is the intended removal of IA from the organic phase. At the same time, the TOA α -methylene resonance near 2.95 ppm evolves from a more structured triplet or multiplet into a broader, more averaged signal as the HCl concentration increases. This is consistent with protonation of the amine and rapid exchange between free TOA, TOA-HCl, and associated ion-paired species, rather than with extensive degradation of the extractant.

Most importantly, the spectra remain largely clean throughout the experiment, which is highly encouraging for solvent recycle. Propyl acetate shows no obvious new resonances over the full time series, indicating that both the diluent and the extractant remain chemically stable under the inhibitory precipitation conditions. The 2-octanone system shows only a very weak additional singlet near 5.0 ppm in the final spectrum, which suggests at most trace side-product formation. The 1-octanol system exhibits a more noticeable new signal near 2.35 ppm at high HCl loading, but this feature appears only in the late stage of the experiment after the system has entered the second plateau.

The fact that new peaks, where present, appear only at high HCl concentration and only after most of the TOA has already been converted to TOA-HCl suggests that impurity formation is limited both kinetically and thermodynamically during most of the process. In the early and intermediate stages, the incoming HCl is preferentially scavenged by free TOA, so the acid is largely consumed in the intended protonation reaction rather than remaining available to catalyse side chemistry. Under these conditions, HCl mainly drives competitive displacement of IA from the TOA-IA complex, while the organic phase is effectively buffered by the remaining free amine. As a result, the concentration of unbound, strongly acidic species capable of promoting solvent degradation or secondary reaction pathways remains low, which explains why the spectra stay largely unchanged over most of the experiment.

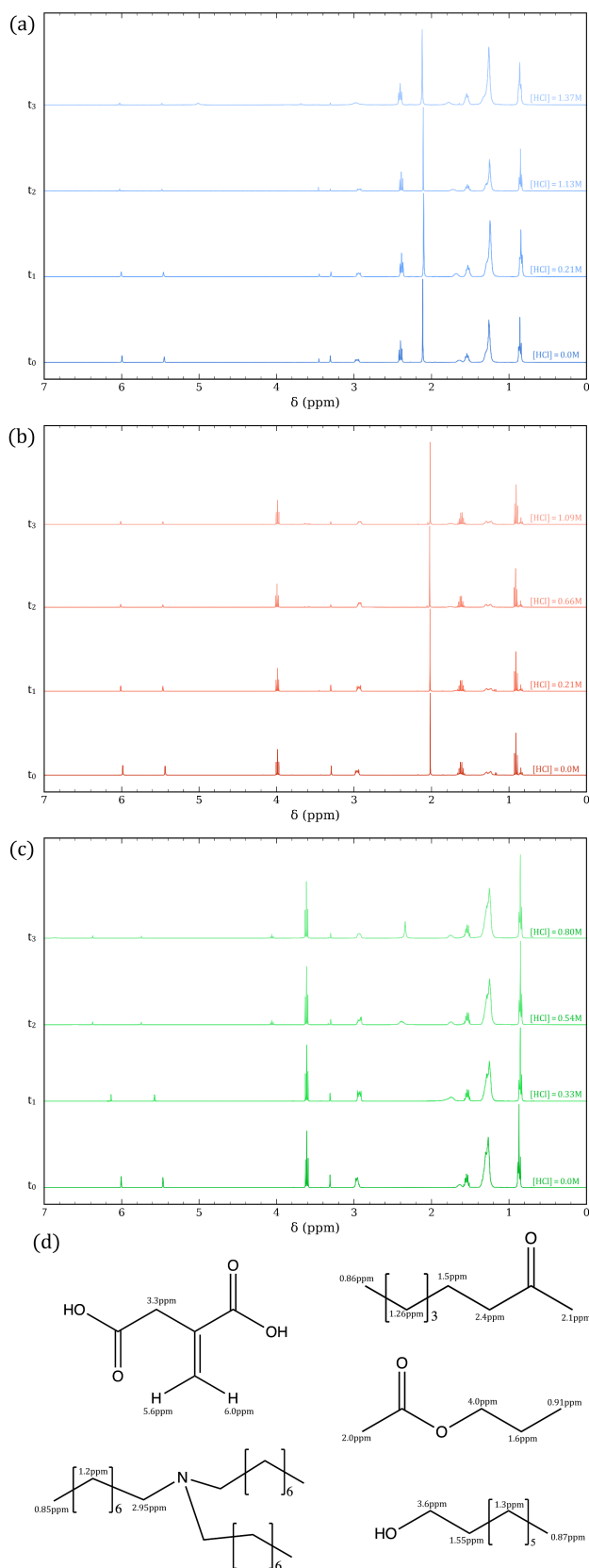


Figure 7: ^1H NMR spectra recorded during inhibitory precipitation in (a) 2-octanone, (b) propyl acetate, and (c) 1-octanol. In each panel, t_0 is the initial loaded organic phase, t_1 is near the end of the first plateau, t_2 is near the end of the linear descent, and t_3 is in the second plateau at the end of the experiment; the corresponding HCl concentrations are indicated beside each spectrum. Panel (d) shows the structures of IA, TOA, and the diluents with approximate proton assignments. The residual CDCl_3 peak is omitted for clarity.

Only in the late stage, once the free TOA concentration has been substantially depleted and the system has entered the second plateau, does this buffering capacity diminish enough for direct acid-catalysed side reactions to become more likely. On this basis, the weak new resonance observed in the 2-octanone system may tentatively be associated with trace enolisation-related chemistry or minor acid-catalysed aldol condensation, while the additional signal in the 1-octanol system may arise from very limited formation of chlorooctane or from esterification between IA and octanol. These assignments remain speculative, since the present 1D spectra do not permit definitive structural identification. Nevertheless, the late appearance and low intensity of these signals indicate that such reactions are secondary and limited in extent. Overall, the spectral evolution remains highly encouraging and suggests that inhibitory precipitation leaves an organic phase that is largely suitable for recycle.

It should also be noted that these samples were collected at 273.15 K. If TOA regeneration were later carried out thermally, impurity formation could become more pronounced and should be reassessed under those conditions. Even in that case, base regeneration with NaOH remains a straightforward alternative route for TOA recovery.

4 Conclusion

This work demonstrates that inhibitory precipitation is a feasible new back-extraction route for the reactive extraction of small dicarboxylic acids, using itaconic acid as a case study. By introducing dry HCl gas directly into the loaded organic phase, the extracting ability of TOA was suppressed through protonation, displacing IA from the acid-amine complex and causing it to precipitate directly as the free acid. The results show that back-extraction can therefore be achieved without an aqueous stripping phase, while simultaneously intensifying the product stream and avoiding the formation of salt waste. This distinguishes inhibitory precipitation from conventional pH-swing stripping and highlights its broader potential as an alternative recovery concept for dicarboxylic acids.

The stoichiometric behaviour of inhibitory precipitation can be interpreted as a reversed solubility problem. As HCl was added, the concentration of free TOA available to solubilise IA progressively decreased, and precipitation began once the remaining free TOA could no longer sustain the initial acid loading. For all three solvent systems, the onset of precipitation was predicted well from the solubility of IA as a function of TOA concentration, showing that the critical HCl dosage is closely linked to the TOA-dependent solubilisation capacity of the organic phase. Differences in the slope of the descending region further showed that the effective stoichiometric response during back-extraction depends on the solvent environment. 1-octanol showed near-ideal agreement with the predicted behaviour, propyl acetate also remained reasonably close, and 2-octanone deviated more strongly, suggesting a more complex redistribution of acid-containing species during protonation. The maximum amount of carboxylic acid that can be back-extracted at a given temperature can likewise be estimated reasonably well from the intrinsic solubility at that temperature, although the experimen-

tally observed residual concentration was consistently somewhat higher than this limiting value.

The purity results further support the practicality of the concept. FTIR confirmed that the precipitated solid was itaconic acid, with no evidence of major impurity formation in the recovered solid. NMR analysis of the organic phase showed that the dominant change during HCl addition was the intended disappearance of IA from solution together with protonation of TOA. Only minor new resonances were observed at high HCl loading, and these appeared only in the late stage of the process after most free TOA had already been converted to TOA-HCl. This indicates that side-product formation is suppressed during the main operating window and suggests that the organic phase remains largely suitable for recycle.

Among the diluents studied, propyl acetate appears to provide the best overall balance between forward extraction and back-extraction performance. It gave the highest extraction efficiency among the solvents investigated, maintained a relatively low viscosity, showed a back-extraction response close to a 1:1 stoichiometry with respect to HCl and itaconic acid, and exhibited the cleanest NMR behaviour during HCl treatment. 2-octanone also performed well in terms of extraction and solvent stability, but its much shallower precipitation slope suggests a less direct stoichiometric response during back-extraction. 1-octanol gave the most ideal precipitation slope, but its substantially higher viscosity, higher intrinsic IA solubility, and somewhat greater tendency toward late-stage side chemistry make it less attractive as an overall process solvent. On this basis, propyl acetate emerges as the most promising solvent for integrating reactive extraction with inhibitory precipitation.

Overall, this study establishes inhibitory precipitation as a credible new route for recovering small dicarboxylic acid from TOA-based reactive extraction systems. The main contribution of the present work is to show that the concept is possible and can function as a new mode of back-extraction, rather than to demonstrate a fully optimised industrial process. In this sense, the use of HCl gas should be viewed primarily as a proof-of-concept implementation of the competitive displacement principle, and not necessarily as the final practical embodiment of the technology. Although effective, HCl gas introduces general handling and process-safety concerns that may limit its attractiveness at scale. Future work should therefore focus not only on cyclic demonstration of extractant regeneration and recycle, but also on identifying alternative agents or strategies that can achieve the same inhibitory effect with lower operational risk.

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